

Syllabus

Components of Food, Diversity in the Living World, Living Creatures – Exploring Their Characteristics, Materials Around Us, Methods of Separation, Air and Water, Motion and Measurement, Temperature and Its Measurement, Exploring Magnets, Light, Beyond Earth.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. Observe the following list of characteristics:
(i) Presence of a backbone,
(ii) Ability to breathe through moist skin,
(iii) Laying eggs in water,
(iv) Ability to live both on land and in water.
To which group of animals do these characteristics collectively apply?
(A) Reptiles (B) Amphibians
(C) Pisces (Fish) (D) Mammals
2. If the zero mark of a ruler is broken, the correct way to measure the length of an object is to
(A) avoid using the ruler
(B) start from any clear marking and subtract the initial reading
(C) estimate the length by eye
(D) reverse the ruler
3. Raincoats are made from materials that do not allow water to pass through. This property is called
(A) Solubility
(B) Permeability
(C) Water resistance
(D) Conductivity
4. A group of students conducted an experiment where they took three test tubes containing starch solution. To Tube A, they added saliva; to Tube B, they added boiled saliva; and Tube C was left as a control. After 20 minutes, they added iodine solution to all three. Which of the following observations and conclusions is correct?
(A) Tube A turns blue-black because saliva contains starch.
(B) Tube B remains colorless because boiling enhances enzyme activity.

(C) Tube A does not change color because saliva breaks down starch into sugars.

(D) Tube C remains colorless as it acts as a neutral agent.

5. Riya measures the length of a classroom blackboard using her handspan and gets 18 handspans. Arjun measures the same blackboard using his handspan and gets 21 handspans. Both measurements are correct. What important idea about measurement does this situation show?

(A) A handspan is not a reliable body part for measuring length

(B) The measurement changes because people have different body sizes

(C) Body parts should not be used at all for measuring

(D) We need fixed and standard units for measurement

6. A mixture contains sand, common salt, and iron filings. Which of the following sequences correctly separates all three components?

A) Use a magnet → add water → filter → evaporate water

B) Dissolve in water → filter → use a magnet → evaporate water

C) Use a magnet → dissolve in water → filter → evaporate water

D) Filter → use a magnet → dissolve in water → evaporate water

7. Consider a desert ecosystem. A specific plant has a thick waxy coating on its stems, reduced leaves in the form of

spines, and roots that grow very deep into the soil. Which characteristic of living organisms is most specifically demonstrated by these features in relation to its environment?

(A) Respiration in extreme conditions.

(B) Adaptation for survival and conservation of water.

(C) Reproduction through vegetative propagation.

(D) Excretion of toxic wastes through spines.

8. A student measures the length of a pencil by placing one end of the pencil at the 2 cm mark of a scale and the other end at the 14.5 cm mark. What is the correct length of the pencil?

(A) 14.5 cm (B) 12.5 cm

(C) 16.5 cm (D) 2 cm

9. When a balloon is taken from a cold place to a warm place, it expands because

(A) Oxygen reacts with nitrogen.

(B) Air molecules move faster and occupy more space.

(C) The balloon produces more air.

(D) Air molecules become heavier.

10. A person is suffering from bleeding gums and takes a long time for wounds to heal. The doctor suggests increasing the intake of citrus fruits and amla. Which of the following explains the logical connection between the symptoms and the remedy?

(A) The person has Scurvy caused by Vitamin C deficiency; citrus fruits are rich in Vitamin C.

(B) The person has Rickets caused by Vitamin D deficiency; citrus fruits help absorb Calcium.

(C) The person has Anemia caused by Iron deficiency; amla helps in blood formation.

(D) The person has Beri-beri caused by Vitamin B1 deficiency; amla restores nerve function.

11. The tip of the second hand of a clock moves continuously and comes back to the same position after every 60 seconds. This motion can best be described as:

(A) Linear motion only

(B) Oscillatory motion only

(C) Circular and periodic motion

(D) Random motion

12. Windows in homes are made of glass because it is (Materials around us)

(A) Flexible and soft

(B) Transparent and allows light in

(C) Heavy and opaque

(D) Conducts electricity

13. An organism 'X' lives in a pond. It takes in food, grows in size, produces waste, and gives rise to young ones of its own kind. However, it does not have a brain or a heart. Organism 'X' is most likely:

(A) A piece of floating wood that grows by absorbing water.

(B) A microscopic living organism like an Amoeba or a simple water plant.

(C) A non-living robot programmed to mimic biological functions.

(D) A dead leaf that is decomposing and breaking into smaller parts.

14. A cyclist moves 200 m east, then 200 m west, and stops at the starting point. Which statement is correct?

(A) Distance is zero, position changed

(B) Distance is zero, no motion occurred

(C) Distance is 400 m, position unchanged

(D) Distance is 200 m, position unchanged

15. Which method is not suitable for separating two liquids with very close boiling points?

(A) Simple distillation

(B) Fractional distillation

(C) Evaporation

(D) Both B and C

16. A student noticed that some animals, like tigers, have eyes in the front of their heads, while others, like deer, have eyes on the sides of their heads. How does this diversity in physical features help them survive?

(A) Side eyes help the tiger focus on its prey accurately.

(B) Front eyes allow the deer to see in all directions to detect danger.

(C) Front eyes provide the tiger with "binocular vision" to judge distance, while side eyes give the deer a wide field of view.

(D) There is no functional difference; it is just a matter of facial structure.

17. Which option correctly matches the object with its type of motion?
- (A) Moving bus on road – Oscillatory motion
 (B) Earth around the Sun – Circular and periodic motion
 (C) Pendulum of a clock – Linear motion
 (D) Spinning top – Linear motion
18. Which of the following is a result of air pollution?
- (A) Cloud formation (B) Acid rain
 (C) Water cycle (D) Condensation
19. **Assertion:** All living organisms show movement, but not all living organisms show locomotion.
Reason: Locomotion is the ability of an organism to move from one place to another, while movement can simply be the motion of a body part.
- (A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
 (B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.
20. During a science experiment, Rahul touches three different objects kept on a table in an air-conditioned room (maintained at 20°C). Object X is a steel spoon; Object Y is a wooden block, and Object Z is a plastic ruler. Even though they have been in the room for hours, Rahul feels that Object X is much colder

than Object Y and Z. Which of the following best explains this sensation?

- (A) The steel spoon has a lower actual temperature than the wooden block and plastic ruler.
 (B) Our sense of touch is unreliable; it measures the rate of heat transfer rather than the actual temperature.
 (C) The wooden block is generating its own heat, making it feel warmer than the steel spoon.
 (D) The plastic ruler is an insulator, so it absorbs heat from the room faster than steel.

21. Rubber or soft plastic is used to make toys because these materials are:

- (A) Hard and brittle
 (B) Flexible and soft
 (C) Transparent
 (D) Conductive

22. **Assertion:** Dietary fibers (roughage) do not provide any nutrients to our body.

Reason: Roughage is an essential part of our food because it helps in the easy passage of stool and prevents constipation.

- (A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.
 (B) Both Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.

23. You are provided with two thermometers: Thermometer A is a standard Clinical Thermometer, and Thermometer B is a Laboratory Thermometer. You need to measure the temperature of boiling water (100°C). Which of the following statements correctly evaluates the suitability of these thermometers for this task?

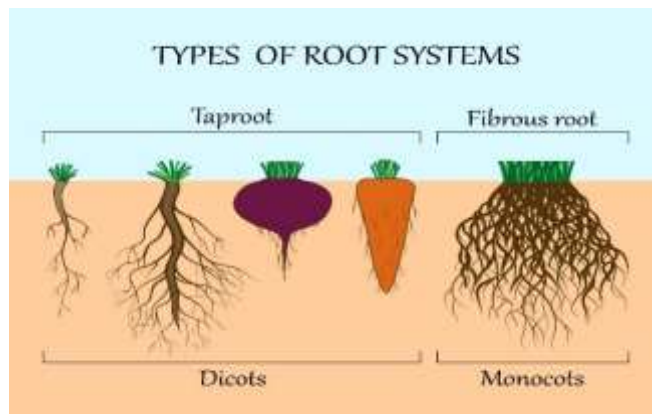
- (A) Both A and B can be used if handled carefully.
- (B) Only Thermometer A can be used because it is more precise.
- (C) Only Thermometer B can be used because Thermometer A has a short range (usually 35°C to 42°C) and might break.
- (D) Neither can be used; you require a digital thermometer for boiling water.

24. A student wants to separate ink from water. Which methods can be used?

1. Evaporation
2. Chromatography
3. Filtration
4. Distillation

- (A) 1 and 3
- (B) 2 only
- (C) 2 and 4
- (D) 1, 2, and 4

25. Look at the two types of root systems, 'X' and 'Y', illustrated below. Which of the following statements correctly identifies the diversity in these plants?



- (A) 'X' is a fibrous root system found in plants like wheat; 'Y' is a taproot system found in pea plants.
- (B) 'X' is a taproot system found in dicots like carrot; 'Y' is a fibrous root system found in monocots like maize.
- (C) 'X' helps the plant survive in deserts; 'Y' is only found in aquatic plants.
- (D) Both root systems are identical in function and are found in the same species of plants.

26. Four students are discussing the proper method to read a laboratory thermometer while measuring the temperature of warm water in a beaker. Who is following the correct procedure?

- Ankit: Takes the thermometer out of the water, holds it horizontally, and reads it.
- Bina: Keeps the thermometer in the water, ensures the bulb does not touch the bottom or sides, and reads it while holding it vertically.
- Chirag: Keeps the thermometer in the water, rests the bulb on the bottom of the beaker for support, and reads it.

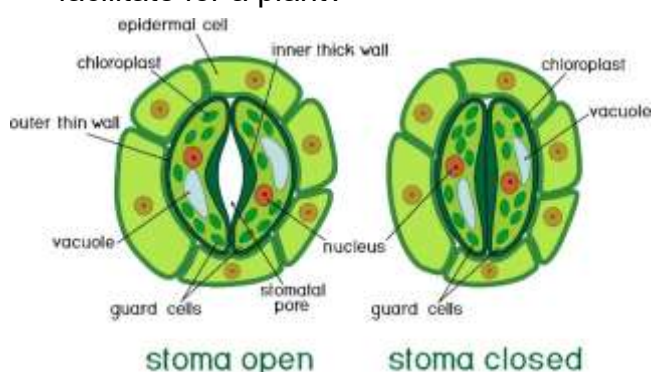
- Diya: Takes the thermometer out, shakes it to settle the mercury, and then reads it.

(A) Ankit (B) Bina
(C) Chirag (D) Diya

27. What will happen to a burning candle placed inside a sealed container after the air is removed

(A) Flame grows brighter
(B) Flame becomes blue
(C) Flame extinguishes
(D) Flame remains the same

28. Observe the diagram of a stomata (tiny pores on a leaf). What are the two main life characteristics that these structures facilitate for a plant?



(A) Reproduction and Movement.
(B) Photosynthesis (Gas exchange) and Transpiration (Excretion of water).
(C) Protection from predators and Absorption of minerals from soil.
(D) Digestion of insects and Storage of starch.

29. Why are thick walls in cold regions made of insulating materials?

(A) They allow heat to escape quickly
(B) They prevent heat from escaping

(C) They are transparent
(D) They conduct heat easily

30. Which mixture can be separated by both evaporation and crystallization, but only crystallization gives pure substance?

(A) Salt + water (B) Sand + water
(C) Sugar + sand (D) Oil + water

31. Consider the scale of a standard Clinical Thermometer which ranges from 35°C to 42°C. If a healthy human body temperature is approximately 37°C, which of the following temperatures would indicate a high fever requiring immediate medical attention?

(A) 36.8°C (B) 37.0°C
(C) 37.4°C (D) 40.2°C

32. Consider the following scenario: A lake is polluted due to dumping of industrial waste. Which effects are most likely?

i) Death of aquatic life
ii) Groundwater contamination
iii) Increase in oxygen in water
iv) Waterborne diseases in humans

(A) i and ii only (B) i, ii, and iv
(C) ii and iii only (D) iii and iv only

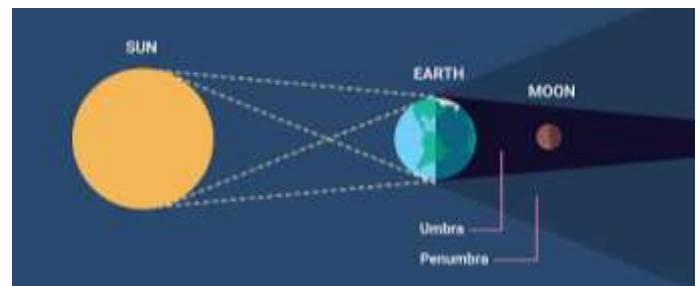
33. A mixture contains iron filings, sawdust, copper shavings, and nickel coins. If you pass a strong bar magnet over this mixture without touching it, which set of materials will be separated from the rest?

(A) Iron filings and copper shavings
(B) Only iron filings
(C) Iron filings and nickel coins
(D) Sawdust and copper shavings

- 34.** To store acids safely, a chemist should use:
- (A) Metal containers
 - (B) Wooden containers
 - (C) Glass containers
 - (D) Rubber containers
- 35.** A refrigerator door remains closed tightly even without a lock. Which application of magnets is responsible for this?
- (A) Finding directions
 - (B) Producing electric current
 - (C) Magnetic strips holding the door
 - (D) Temporary magnetization
- 36.** Consider a bird flying high up in the sky. The sun is shining brightly overhead. Why do we not see the bird's shadow on the ground?
- (A) The bird is transparent to sunlight.
 - (B) The bird is too small to block any sunlight.
 - (C) The screen (ground) is too far away from the object (bird), so the shadow becomes too faint and disappears (umbra vanishes).
 - (D) Shadows can form in open air.
- 37.** Which method cannot be used to separate a homogeneous mixture?
- (A) Distillation
 - (B) Chromatography
 - (C) Decantation
 - (D) Fractional distillation
- 38.** You are given a magnet with no markings. You also have a magnetic compass. When you bring one end of the magnet near the North-seeking pole (Red tip) of the compass needle, the needle turns AWAY. What pole is that end of the magnet?
- (A) North Pole
 - (B) South Pole
 - (C) Neutral Pole
 - (D) Iron Pole
- 39.** Which of the following is an example of conservation of water?
- (A) Letting taps run while brushing
 - (B) Planting trees
 - (C) Wasting water in ponds
 - (D) Using water for washing roads unnecessarily
- 40.** Why do magnets lose their strength when they are heated, hammered, or dropped repeatedly?
- (A) Magnetic material melts on heating
 - (B) The alignment of magnetic domains gets disturbed
 - (C) Poles disappear permanently
 - (D) Air enters the magnet and weakens it
- 41.** Plastic bottles are preferred over glass bottles for soft drinks because plastic is:
- (A) Heavier and transparent
 - (B) Lightweight, unbreakable, and inexpensive
 - (C) Transparent but brittle
 - (D) Conductive and hard
- 42.** Planet 'X' is known as the "Red Planet" because of the iron oxide (rust) on its surface. It has two small, irregular moons named Phobos and Deimos. Which planet is this?
- (A) Venus
 - (B) Jupiter
 - (C) Mars
 - (D) Mercury

- 43.** Which separation method is most effective for separating two liquids that have very similar boiling points?
- (A) Simple distillation
(B) Fractional distillation
(C) Evaporation
(D) Decantation
- 44.** You look up at the night sky and see a group of stars forming the shape of a hunter with a belt of three distinct bright stars. Which constellation are you looking at?
- (A) Ursa Major (B) Orion
(C) Cassiopeia (D) Leo
- 45.** Which planet in our solar system would float if you could find a bathtub big enough to put it in?
- (A) Earth (B) Mars
(C) Saturn (D) Uranus
- 46.** Why are more stars visible on a night when there is no Moon?
- (A) Stars shine brighter
(B) Moon blocks the stars
(C) Moonlight makes the sky bright
(D) Stars appear only at night
- 47.** Which planet takes about 30 years to complete one round around the Sun and has beautiful rings?
- (A) Jupiter (B) Saturn
(C) Uranus (D) Neptune
- 48.** An object placed in a completely dark room cannot be seen because—
- (A) the object absorbs light
(B) light is not present
(C) the object is too small
(D) eyes stop working

- 49.** Why should we never look directly at a Solar Eclipse with naked eyes?
- (A) It is bad luck.
(B) The Moon's dust can enter our eyes.
(C) The intense UV radiation from the Sun can permanently damage the retina.
(D) The sky is too dark to see anything.
- 50.** Imagine you are on the Moon. You look up and see the Earth passing directly in front of the Sun, blocking it. What event are people on Earth experiencing at that same moment?



- (A) A Lunar Eclipse (B) A Solar Eclipse
(C) A New Moon (D) Sunrise

KEY

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|-------|-------|-------|-------|-------|-------|-------|
| 1) B | 2) B | 3) C | 4) C | 5) D | 6) C | 7) B |
| 8) B | 9) B | 10) A | 11) C | 12) B | 13) B | 14) C |
| 15) A | 16) C | 17) B | 18) B | 19) A | 20) B | 21) B |
| 22) B | 23) C | 24) B | 25) B | 26) B | 27) C | 28) B |
| 29) B | 30) A | 31) D | 32) B | 33) C | 34) C | 35) C |
| 36) C | 37) C | 38) A | 39) B | 40) B | 41) B | 42) C |
| 43) B | 44) B | 45) C | 46) C | 47) B | 48) B | 49) C |
| 50) A | | | | | | |